INCH-POUND

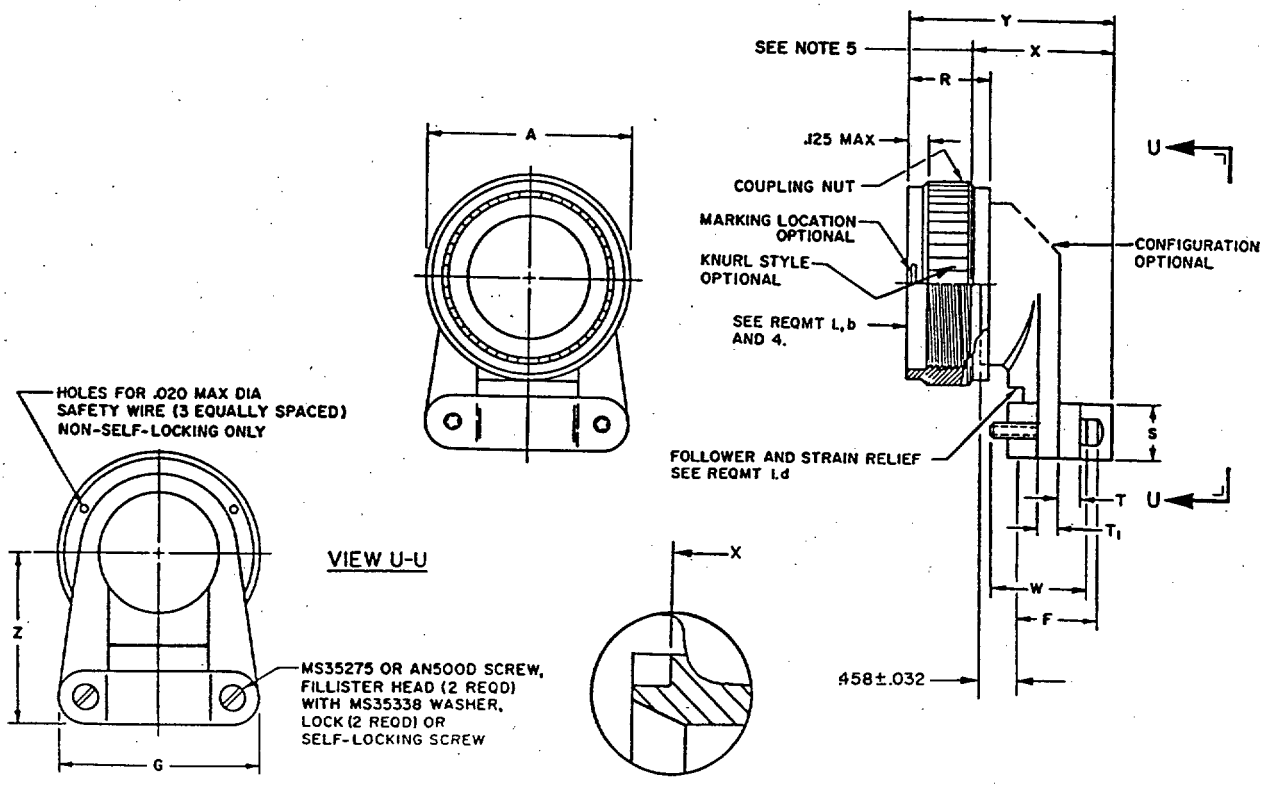
MIL-C-85049/51C
30 November 1990
SUPERSEDING
MIL-C-0085049/51B(USAF)
28 April 1989
MIL-C-0085049/51A(USAF)
21 November 1986
MIL-C-85049/51
16 January 1980
MS3418E
10 May 1982

# MILITARY SPECIFICATION SHEET

## CONNECTOR ACCESSORIES, ELECTRICAL, STRAIN RELIEF, NONENVIRONMENTAL, 90°, SELF-LOCKING AND NON-SELF-LOCKING CATEGORY 4B (FOR MIL-C-5015 CRIMP, MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3, AND MIL-C-83723 SERIES III CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.




NON-SELF-LOCKING AND SELF-LOCKING

FIGURE 1. Configuration and dimensions.

AMSC N/A
FSC 5935

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

TABLE I. Shell size, dash numbers and dimensions.

| Dash No. | For Connector Shell, Size (Ref) | | | | Non-Self-Locking | | | | | | Self-Locking | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MIL-C-81703 Series 3 | MIL-C-26482 Series 2 | MIL-C-5015 Crimp | MIL-C-83723 Series III | A Max Dia | R Max | Fillister Head Screw | | X Max | Y Max | A Max Dia | R Max | X Max | Y Max |
| | | | | | | | AN500D Dash No. | MS35275 Dash No. | | | | | | |
| -3 | 3 | | | | .669 | .540 | 6-8 | 230 | .78 | 1.101 | | | | |
| -8 | | 8 | 8S | 8 | .617 | ↑ | 6-8 | 230 | .78 | 1.101 | .885 | .710 | .93 | 1.257 |
| -10 | | 10 | 10S&10SL | 10 | .734 | | 6-8 | 230 | .86 | 1.183 | 1.010 | ↑ | 1.02 | 1.339 |
| -12 | 7 | 12 | 12S&12 | 12 | .858 | | 6-10 | 231 | 1.05 | 1.376 | 1.135 | | 1.21 | 1.532 |
| -14 | 12 | 14 | 14S&14 | 14 | .984 | | 6-10 | 231 | 1.11 | 1.436 | 1.260 | | 1.27 | 1.592 |
| -16 | 19 | 16 | 16S&16 | 16 | 1.112 | | 6-10 | 231 | 1.26 | 1.585 | 1.385 | | 1.42 | 1.741 |
| -18 | 27 | 18 | 18 | 18 | 1.218 | | 8-12 | 247 | 1.37 | 1.697 | 1.510 | | 1.53 | 1.853 |
| -20 | 37 | 20 | 20 | 20 | 1.345 | | 8-14 | 248 | 1.50 | 1.822 | 1.635 | | 1.65 | 1.978 |
| -22 | | 22 | 22 | 22 | 1.468 | ↓ | 8-16 | 249 | 1.62 | 1.947 | 1.760 | ↓ | 1.78 | 2.108 |
| -24 | | 24 | 24 | 24 | 1.593 | .540 | 8-18 | N/A | 1.75 | 2.072 | 1.885 | .710 | 1.90 | 2.228 |
| -28 | | | 28 | | 1.969 | .702 | 8-20 | 250 | 1.89 | 2.372 | 2.135 | .890 | 2.20 | 2.528 |
| -32 | | | 32 | | 2.219 | ↑ | 416-20 | 284 | 1.95 | 2.435 | 2.395 | ↑ | 2.27 | 2.591 |
| -36 | | | 36 | | 2.469 | | 416-24 | 285 | 2.07 | 2.560 | 2.635 | | 2.39 | 2.716 |
| -40 | | | 40 | | 2.719 | | 416-24 | 285 | 2.20 | 2.685 | 2.885 | | 2.52 | 2.841 |
| -44 | | | 44 | | 2.969 | ↓ | 416-28 | 286 | 2.45 | 2.935 | 3.135 | ↓ | 2.77 | 3.091 |
| -48 | | | 48 | | 3.219 | .702 | 416-32 | 297 | 2.57 | 3.060 | 3.385 | .890 | 2.89 | 3.216 |
| -61 | 61 | | | | 1.653 | .540 | 8-18 | N/A | 1.75 | 2.072 | | | | |

N/A = Not Available



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

TABLE I. Shell size, dash numbers and dimensions (continued).

| Non-Self-Locking and Self-Locking | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Dash No. | F (see Note 5) Wire Bundle Accommodation Range | | G ±.020 | S ±.020 | T T1 ±.015 | Z ±.062 | MS35338 Washer, Lock Dash Number | W +.032 Screw Length | Screw Size |
| | Max | Min | | | | | | | |
| -3 1/ | .204 | .125 | .732 | .250 | .125 | .777 | 136 | .500 | 6-32 |
| -8 | .204 | .125 | .732 | ↑ | ↑ | .746 | ↑ | .500 | ↑ |
| -10 | .286 | .187 | .812 | ↓ | ↓ | .805 | ↓ | .500 | ↓ |
| -12 | .416 | .291 | .968 | ↓ | ↓ | .867 | ↓ | .625 | ↓ |
| -14 | .476 | .351 | 1.026 | ↓ | ↓ | .930 | ↓ | .625 | ↓ |
| -16 | .626 | .501 | 1.299 | .250 | .125 | .994 | 136 | .625 | 6-32 |
| -18 | .706 | .518 | 1.431 | .375 | .156 | 1.171 | 137 | .750 | 8-32 |
| -20 | .831 | .581 | 1.537 | ↑ | ↑ | 1.234 | ↑ | .875 | ↑ |
| -22 | .956 | .644 | 1.633 | ↓ | ↓ | 1.296 | ↓ | 1.000 | ↓ |
| -24 | 1.081 | .706 | 1.755 | .375 | .156 | 1.358 | ↓ | 1.125 | ↓ |
| -28 | 1.187 | .750 | 2.000 | .500 | .188 | 1.572 | 137 | 1.250 | 8-32 |
| -32 | 1.250 | .875 | 2.366 | ↑ | ↑ | 1.797 | 139 | 1.250 | 1/4-20 |
| -36 | 1.375 | .938 | 2.476 | ↓ | ↓ | 1.922 | ↑ | 1.500 | ↑ |
| -40 | 1.500 | .938 | 2.546 | .500 | ↓ | 2.047 | ↓ | 1.500 | ↓ |
| -44 | 1.750 | 1.188 | 2.840 | .625 | ↓ | 2.296 | ↓ | 1.750 | ↓ |
| -48 | 1.875 | 1.312 | 3.324 | .625 | .188 | 2.421 | 139 | 2.000 | 1/4-20 |
| -61 1/ | 1.081 | .706 | 1.755 | .375 | .156 | 1.388 | 137 | 1.125 | 8-32 |

1/ Not available on self-locking configuration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

| Inch | mm | Inch | mm | Inch | mm | Inch | mm |
|---|---|---|---|---|---|---|---|
| 0.015 | 0.380 | 0.885 | 22.470 | 1.420 | 36.060 | 2.135 | 54.220 |
| 0.020 | 0.500 | 0.890 | 22.600 | 1.431 | 36.340 | 2.200 | 55.880 |
| 0.032 | 0.810 | 0.930 | 23.620 | 1.436 | 36.470 | 2.219 | 56.360 |
| 0.062 | 1.570 | 0.938 | 23.820 | 1.468 | 37.280 | 2.228 | 56.590 |
| 0.125 | 3.170 | 0.956 | 24.280 | 1.500 | 38.100 | 2.270 | 57.650 |
| 0.156 | 3.960 | 0.968 | 24.580 | 1.510 | 38.350 | 2.296 | 58.310 |
| 0.187 | 4.740 | 0.984 | 24.990 | 1.530 | 38.860 | 2.366 | 60.090 |
| 0.188 | 4.770 | 0.994 | 25.240 | 1.532 | 38.910 | 2.372 | 60.240 |
| 0.204 | 5.180 | 1.000 | 25.400 | 1.537 | 39.030 | 2.390 | 60.700 |
| 0.250 | 6.350 | 1.010 | 25.650 | 1.572 | 39.920 | 2.395 | 60.830 |
| 0.286 | 7.260 | 1.020 | 25.900 | 1.585 | 40.250 | 2.421 | 61.490 |
| 0.291 | 7.390 | 1.026 | 26.060 | 1.592 | 40.430 | 2.435 | 61.840 |
| 0.351 | 8.910 | 1.050 | 26.670 | 1.593 | 40.460 | 2.450 | 62.230 |
| 0.375 | 9.520 | 1.081 | 27.450 | 1.620 | 41.140 | 2.469 | 62.710 |
| 0.416 | 10.560 | 1.087 | 27.600 | 1.633 | 41.470 | 2.476 | 62.890 |
| 0.476 | 12.090 | 1.100 | 27.940 | 1.635 | 41.520 | 2.520 | 64.000 |
| 0.500 | 12.700 | 1.110 | 28.190 | 1.650 | 41.910 | 2.528 | 64.210 |
| 0.501 | 12.720 | 1.112 | 28.240 | 1.653 | 41.980 | 2.546 | 64.666 |
| 0.518 | 13.150 | 1.125 | 28.570 | 1.697 | 43.100 | 2.560 | 65.020 |
| 0.540 | 13.710 | 1.126 | 28.600 | 1.700 | 43.180 | 2.570 | 65.270 |
| 0.581 | 14.750 | 1.135 | 28.820 | 1.741 | 44.220 | 2.591 | 65.810 |
| 0.617 | 15.670 | 1.137 | 28.870 | 1.750 | 44.450 | 2.635 | 66.920 |
| 0.625 | 15.870 | 1.171 | 29.740 | 1.755 | 44.570 | 2.685 | 68.190 |
| 0.626 | 15.900 | 1.183 | 30.040 | 1.760 | 44.700 | 2.716 | 68.980 |
| 0.644 | 16.350 | 1.188 | 30.170 | 1.797 | 45.640 | 2.719 | 69.060 |
| 0.669 | 16.990 | 1.210 | 30.730 | 1.822 | 46.270 | 2.770 | 70.350 |
| 0.702 | 17.830 | 1.218 | 30.930 | 1.853 | 47.060 | 2.840 | 72.130 |
| 0.706 | 17.430 | 1.234 | 31.340 | 1.875 | 47.620 | 2.841 | 72.160 |
| 0.730 | 18.540 | 1.250 | 31.750 | 1.885 | 47.870 | 2.885 | 73.270 |
| 0.734 | 18.640 | 1.257 | 31.920 | 1.890 | 48.000 | 2.890 | 73.400 |
| 0.746 | 18.940 | 1.270 | 32.250 | 1.900 | 48.260 | 2.935 | 74.540 |
| 0.750 | 19.050 | 1.246 | 32.910 | 1.922 | 48.810 | 2.969 | 75.410 |
| 0.777 | 19.730 | 1.299 | 32.990 | 1.947 | 49.450 | 2.970 | 75.430 |
| 0.780 | 19.810 | 1.312 | 33.320 | 1.950 | 49.530 | 3.011 | 76.470 |
| 0.805 | 20.440 | 1.339 | 34.010 | 1.969 | 50.010 | 3.060 | 77.720 |
| 0.812 | 20.620 | 1.345 | 34.160 | 1.978 | 50.240 | 3.135 | 79.620 |
| 0.831 | 21.100 | 1.358 | 34.490 | 2.000 | 50.800 | 3.216 | 81.680 |
| 0.858 | 21.79 | 1.375 | 34.920 | 2.047 | 51.990 | 3.219 | 81.760 |
| 0.860 | 21.840 | 1.376 | 34.950 | 2.070 | 52.570 | 3.324 | 84.420 |
| 0.867 | 22.020 | 1.385 | 35.170 | 2.072 | 52.620 | 3.385 | 85.976 |
| 0.875 | 22.220 | 1.388 | 35.250 | 2.108 | 53.540 | | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Design and construction:

   a. Dimensions and configuration: See figure 1 and table I.

   b. Interface dimensions shall be in accordance with MIL-C-85049 figure 4.

   c. This clamp will not accommodate connectors using size 8, 4 and 0 contacts.

   d. Accessory consists of a coupling nut and follower. The coupling nut shall be captivated to and free to rotate on the follower.

   e. Clamps shall have no protrusions or sharp edges which may pinch cables.

2. Material and finish: See Table II.

TABLE II. Material and finish.

| Figure | Material | Finish |
|---|---|---|
| 1 & 2 | Aluminum alloy in accordance with MIL-C-85049 | 1/ N, 2/ W<br>A |

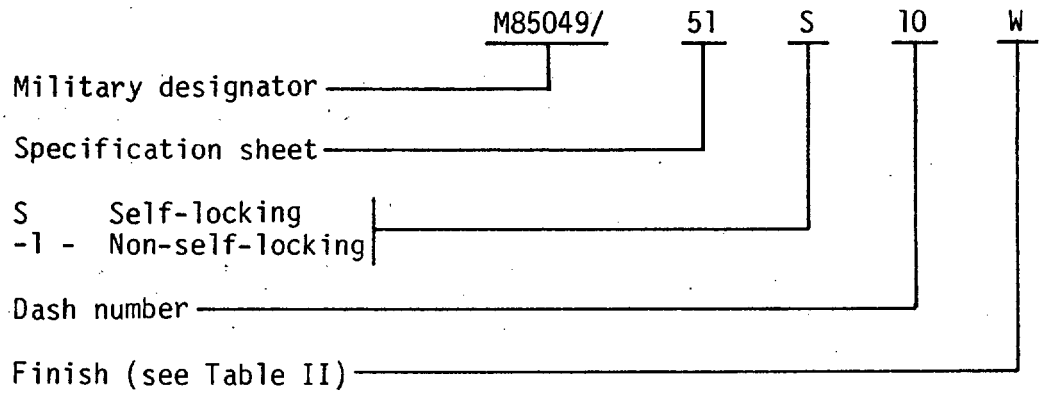
1/ Not for Navy use. Restricted to Air Force Space applications only.
2/ W finish is not for use in space application.

3. Qualification:

   Non-self-locking and self-locking - See MIL-C-85049, category 4B.

   Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

4. Part number: See example below:






Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

5. Supersession data:

MS3418 - * * M85049/51-1 - * *

Dash Number

Finish

| Old | New |
|---|---|
| A | W |
| C | W |
| G | A |
| N | N |

M83723/15A * * M85049/51-1 - * *

Shell Size Dash Number

Class Finish

| Old | New |
|---|---|
| A | W |
| R | N |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.
3. Dimensions apply after plating.
4. "Wire bundle accommodation range" dimension is defined as the envelope area of the wire bundle. This dimension is not meant to define the clamp hardware limits.
5. Measured from the root of the tooth to the back of the coupling nut.

Revision letters are not used to denote changes due to the extensiveness of the changes.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 99
DLA - ES
NASA - JFK, LRC, GSF

Preparing activity:
Navy - AS
Project No. 5935-3734-02



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:12Z
Check the source to verify that this is the current version before use.